Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 1000 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*32*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *32.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1229*	*0.04012*
Peak wave period T_P (s):	*1.882*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.785*	*1.76*
Mean wave period $T_{0,1}$ (s):	*1.603*	*1.287*
Mean wave period $T_{0,2}$ (s):	*1.485*	*1.066*
Spectral width (Broadness):	*0.7575*	*0.8389*
Spectral width (Narrowness):	*0.406*	*0.6774*

Time Domain Analysis

Overall amplitude reflection coefficient: *32.6%*

	Incident	**Reflected**
Number of waves:	1003	1137
Mean wave height H_m (m):	*0.0785*	*0.0259*
Mean wave period T_m (s):	*1.658*	*1.461*
Sig. wave height H_s (m):	*0.1209*	*0.04275*
$T_{H_{1/3}}$ (s):	*1.903*	*1.852*
H_{max} (m):	*0.1923*	*0.09527*
T_{Hmax} (s):	*2.16*	*2.013*
$H_{1/10}$ (m):	*0.1479*	*0.05693*
$H_{1/50}$ (m):	*0.1739*	*0.0707*
$H_{1/100}$ (m)	*0.1803*	*0.07621*
$H_{1/250}$ (m):	*0.185*	*0.08245*
$H_{0.1\%}$ (m):	*0.1923*	*0.09446*
$H_{1\%}$ (m):	*0.1744*	*0.0701*
$H_{2\%}$ (m):	*0.1644*	*0.06286*
$H_{10\%}$ (m)	*0.1289*	*0.04691*
Groupiness factor GF:	*1.048*	*1.25*
Skewness b1:	*0.6865*	*0.3022*
Kurtosis b2:	*3.263*	*3.866*

Department of Civil Engineering

Aalborg University

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 1 Overtop.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.5%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1348*	*0.04655*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.838*	*1.814*
Mean wave period $T_{0,1}$ (s):	*1.62*	*1.294*
Mean wave period $T_{0,2}$ (s):	*1.482*	*1.056*
Spectral width (Broadness):	*0.7858*	*0.8474*
Spectral width (Narrowness):	*0.4414*	*0.7085*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.5%*

	Incident	**Reflected**
Number of waves:	1027	1174
Mean wave height H_m (m):	*0.08863*	*0.03119*
Mean wave period T_m (s):	*1.743*	*1.526*
Sig. wave height H_s (m):	*0.1352*	*0.05078*
$T_{H_{1/3}}$ (s):	*1.992*	*1.946*
H_{max} (m):	*0.2254*	*0.09417*
T_{Hmax} (s):	*1.725*	*1.803*
$H_{1/10}$ (m):	*0.1651*	*0.06415*
$H_{1/50}$ (m):	*0.1964*	*0.0766*
$H_{1/100}$ (m)	*0.2058*	*0.08151*
$H_{1/250}$ (m):	*0.2144*	*0.08751*
$H_{0.1\%}$ (m):	*0.2251*	*0.09341*
$H_{1\%}$ (m):	*0.1933*	*0.07459*
$H_{2\%}$ (m):	*0.1785*	*0.06935*
$H_{10\%}$ (m)	*0.1441*	*0.05541*
Groupiness factor GF:	*1.066*	*1.173*
Skewness b1:	*0.7956*	*0.2161*
Kurtosis b2:	*3.4*	*3.533*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 1200 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.227 Hz*
Freq. upper bound:	*4.531 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*42*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *37%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1484*	*0.05494*
Peak wave period T_P (s):	*2.207*	*2.207*
Energy wave period $T_{-1,0}$ (s):	*1.898*	*1.901*
Mean wave period $T_{0,1}$ (s):	*1.643*	*1.305*
Mean wave period $T_{0,2}$ (s):	*1.485*	*1.048*
Spectral width (Broadness):	*0.8105*	*0.8521*
Spectral width (Narrowness):	*0.4739*	*0.7415*

Time Domain Analysis

Overall amplitude reflection coefficient: *37.1%*

	Incident	**Reflected**
Number of waves:	1028	1161
Mean wave height H_m (m):	*0.09754*	*0.03733*
Mean wave period T_m (s):	*1.797*	*1.591*
Sig. wave height H_s (m):	*0.1496*	*0.05872*
$T_{H_{1/3}}$ (s):	*2.052*	*2.002*
H_{max} (m):	*0.2303*	*0.09694*
T_{Hmax} (s):	*2.027*	*1.869*
$H_{1/10}$ (m):	*0.1795*	*0.07179*
$H_{1/50}$ (m):	*0.2097*	*0.08403*
$H_{1/100}$ (m)	*0.218*	*0.08821*
$H_{1/250}$ (m):	*0.225*	*0.0942*
$H_{0.1\%}$ (m):	*0.2302*	*0.09676*
$H_{1\%}$ (m):	*0.2076*	*0.08187*
$H_{2\%}$ (m):	*0.198*	*0.07756*
$H_{10\%}$ (m)	*0.1596*	*0.06342*
Groupiness factor GF:	*1.061*	*1.101*
Skewness b1:	*0.8629*	*0.09742*
Kurtosis b2:	*3.436*	*3.177*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 1500 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *38.5%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1597*	*0.06156*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.93*	*1.946*
Mean wave period $T_{0,1}$ (s):	*1.642*	*1.289*
Mean wave period $T_{0,2}$ (s):	*1.466*	*1.019*
Spectral width (Broadness):	*0.8305*	*0.8567*
Spectral width (Narrowness):	*0.5045*	*0.7755*

Time Domain Analysis

Overall amplitude reflection coefficient: *38.7%*

	Incident	**Reflected**
Number of waves:	1021	1126
Mean wave height H_m (m):	*0.1073*	*0.04339*
Mean wave period T_m (s):	*1.864*	*1.69*
Sig. wave height H_s (m):	*0.1619*	*0.06647*
$T_{H_{1/3}}$ (s):	*2.133*	*2.084*
H_{max} (m):	*0.2377*	*0.1076*
T_{Hmax} (s):	*2.149*	*2.516*
$H_{1/10}$ (m):	*0.1901*	*0.08025*
$H_{1/50}$ (m):	*0.213*	*0.09335*
$H_{1/100}$ (m)	*0.2212*	*0.09746*
$H_{1/250}$ (m):	*0.2311*	*0.1025*
$H_{0.1\%}$ (m):	*0.2376*	*0.107*
$H_{1\%}$ (m):	*0.2106*	*0.09189*
$H_{2\%}$ (m):	*0.1998*	*0.08695*
$H_{10\%}$ (m)	*0.1743*	*0.07087*
Groupiness factor GF:	*1.05*	*1.072*
Skewness b1:	*0.9031*	*-0.02488*
Kurtosis b2:	*3.408*	*3.093*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 1000 (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.203 Hz*
Freq. upper bound:	*4.063 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*47*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *40.2%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1665*	*0.06691*
Peak wave period T_P (s):	*2.462*	*2.462*
Energy wave period $T_{-1,0}$ (s):	*1.971*	*2.031*
Mean wave period $T_{0,1}$ (s):	*1.651*	*1.326*
Mean wave period $T_{0,2}$ (s):	*1.464*	*1.039*
Spectral width (Broadness):	*0.8313*	*0.8563*
Spectral width (Narrowness):	*0.5217*	*0.7928*

Time Domain Analysis

Overall amplitude reflection coefficient: *40.3%*

	Incident	Reflected
Number of waves:	1027	1121
Mean wave height H_m (m):	*0.1129*	*0.04696*
Mean wave period T_m (s):	*1.915*	*1.754*
Sig. wave height H_s (m):	*0.1691*	*0.07064*
$T_{H_{1/3}}$ (s):	*2.196*	*2.176*
H_{max} (m):	*0.2448*	*0.1128*
T_{Hmax} (s):	*2.366*	*2.138*
$H_{1/10}$ (m):	*0.1942*	*0.08389*
$H_{1/50}$ (m):	*0.2154*	*0.09754*
$H_{1/100}$ (m)	*0.2231*	*0.1035*
$H_{1/250}$ (m):	*0.2342*	*0.1085*
$H_{0.1\%}$ (m):	*0.2446*	*0.1125*
$H_{1\%}$ (m):	*0.2138*	*0.097*
$H_{2\%}$ (m):	*0.2023*	*0.08824*
$H_{10\%}$ (m)	*0.1792*	*0.07483*
Groupiness factor GF:	*1.041*	*1.011*
Skewness b1:	*0.9584*	*-0.1044*
Kurtosis b2:	*3.419*	*2.91*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 60% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *21.9%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08267*	*0.01813*
Peak wave period T_P (s):	*1.524*	*1.6*
Energy wave period $T_{-1,0}$ (s):	*1.438*	*1.326*
Mean wave period $T_{0,1}$ (s):	*1.36*	*1.057*
Mean wave period $T_{0,2}$ (s):	*1.31*	*0.9244*
Spectral width (Broadness):	*0.5992*	*0.7843*
Spectral width (Narrowness):	*0.2795*	*0.5543*

Time Domain Analysis

Overall amplitude reflection coefficient: *21.9%*

	Incident	**Reflected**
Number of waves:	977	1251
Mean wave height H_m (m):	*0.0517*	*0.01113*
Mean wave period T_m (s):	*1.342*	*1.048*
Sig. wave height H_s (m):	*0.08054*	*0.01781*
$T_{H_{1/3}}$ (s):	*1.48*	*1.344*
H_{max} (m):	*0.133*	*0.03084*
T_{Hmax} (s):	*1.51*	*1.643*
$H_{1/10}$ (m):	*0.09803*	*0.02295*
$H_{1/50}$ (m):	*0.1159*	*0.02723*
$H_{1/100}$ (m)	*0.1224*	*0.02839*
$H_{1/250}$ (m):	*0.129*	*0.02967*
$H_{0.1\%}$ (m):	*Not enough data*	*0.03081*
$H_{1\%}$ (m):	*0.1136*	*0.02688*
$H_{2\%}$ (m):	*0.1078*	*0.02557*
$H_{10\%}$ (m)	*0.08543*	*0.01916*
Groupiness factor GF:	*0.9682*	*1.111*
Skewness b1:	*0.3276*	*0.1101*
Kurtosis b2:	*2.928*	*3.34*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-5 SWL Berm 2% - Test 3.experiment - 80% (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *26.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1083*	*0.029*
Peak wave period T_P (s):	*1.73*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.632*	*1.542*
Mean wave period $T_{0,1}$ (s):	*1.508*	*1.174*
Mean wave period $T_{0,2}$ (s):	*1.426*	*0.9962*
Spectral width (Broadness):	*0.7009*	*0.819*
Spectral width (Narrowness):	*0.3449*	*0.6238*

Time Domain Analysis

Overall amplitude reflection coefficient: *26.7%*

	Incident	**Reflected**
Number of waves:	983	1257
Mean wave height H_m (m):	*0.06782*	*0.01762*
Mean wave period T_m (s):	*1.513*	*1.183*
Sig. wave height H_s (m):	*0.1042*	*0.02827*
$T_{H_{1/3}}$ (s):	*1.691*	*1.58*
H_{max} (m):	*0.1775*	*0.07101*
T_{Hmax} (s):	*1.594*	*1.479*
$H_{1/10}$ (m):	*0.1279*	*0.03763*
$H_{1/50}$ (m):	*0.1516*	*0.05112*
$H_{1/100}$ (m)	*0.1597*	*0.05708*
$H_{1/250}$ (m):	*0.168*	*0.06228*
$H_{0.1\%}$ (m):	*Not enough data*	*0.06881*
$H_{1\%}$ (m):	*0.151*	*0.05166*
$H_{2\%}$ (m):	*0.1403*	*0.04179*
$H_{10\%}$ (m)	*0.1102*	*0.0301*
Groupiness factor GF:	*0.9981*	*1.26*
Skewness b1:	*0.525*	*0.3497*
Kurtosis b2:	*3.025*	*3.981*

Department of Civil Engineering

Aalborg University

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Drawn by	
Date	*11. mrt 2026*
Remarks	
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